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REVIEW

Green Technology Innovations for Carbon Footprint Reduction in the Restaurant Industry: A Systematic Review

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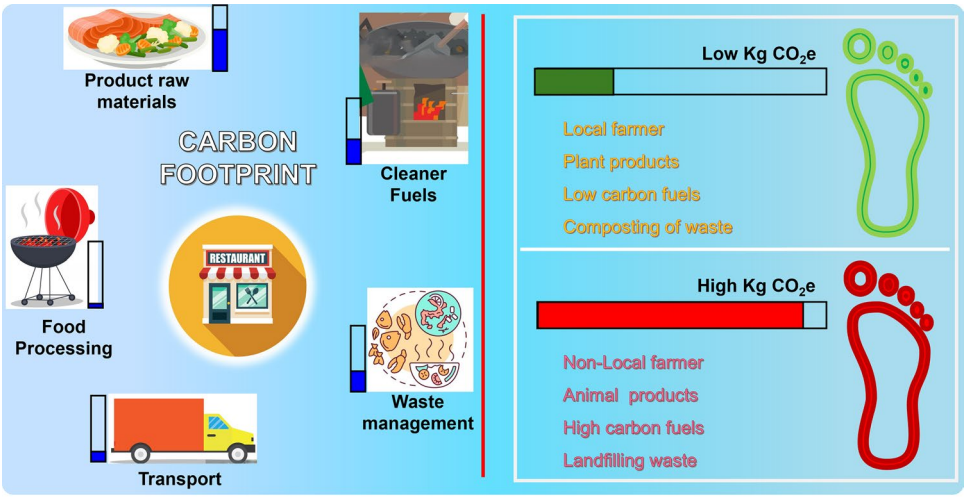
Abstract

Restaurants contribute a significant carbon footprint (CF), with the food service sector responsible for 18% of food-related global CF. This impact arises from energy-intensive processes, carbon-intensive ingredient sourcing, inefficient waste management, and upstream and downstream supply chain activities. Notably, commercial kitchens exhibit carbon footprints 2–5 times greater than those of other commercial spaces. This review synthesizes recent empirical and theoretical literature on carbon footprint reduction strategies in restaurants, with emphasis on the application of life cycle assessment (LCA), IPCC methodologies, and material flow analysis (MFA) across diverse regional contexts. Green technologies such as the integration of solar, biogas, and biodiesel systems have been shown to reduce GHG emissions by 20–75%, while composting and food waste valorization can mitigate methane emissions by up to 30%. Menu-based interventions, including the substitution of beef with plant-based proteins and regionally sourced ingredients, offer substantial emissions reductions at the meal level. Despite these advances, the review identifies persistent challenges, including data inconsistency, underreporting of Scope 3 emissions, and limited adoption of standardized carbon accounting frameworks such as ISO 14064, PAS 2050. The findings highlight the need for harmonized assessment tools, integration of digital monitoring technologies, and cross-sector collaboration to enable consistent benchmarking and guide the sector toward carbon neutrality. This review provides a roadmap for advancing low-carbon transitions through operational, behavioral, and policy-driven interventions in the global restaurant industry.

Graphical Abstract



Keywords Restaurant sustainability · Carbon emissions · Green technology · Waste reduction · Food waste management

Abbreviations

AI	Artificial intelligence
AI	Artificial intelligence

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CF	Carbon footprint
CO_2e	Carbon dioxide equivalent
GHG	Greenhouse gas
GRASS	Green Restaurants Assessment
GS	Green Seal Standard for Food Services
GS-46	Green Seal Standard for Food Services
HFCs	Hydrofluorocarbons
IoT	Internet of Things
IoT	Internet of Things
IPCC	Intergovernmental Panel on Climate Change
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization
KPIs	Key Performance Indicators
LCA	Life cycle assessment
LPG	Liquefied petroleum gas
MFA	Material flow analysis
MFA	Material flow analysis
PAS	Publicly available specification 2050
SDGs	United Nations Sustainable Development Goals

1 Introduction

Amidst the urgency to counter climate change by intensifying focus on greenhouse gas (GHG) reduction across industries, the restaurant sector in the global food system, although a significant contributor, is often overlooked. The entire food sector is estimated to contribute to more than 34% of global GHG emissions, with food service operations alone accounting for approximately 18% of food-related carbon footprint (CF) (Bettencourt et al., 2025; Torres et al. 2022). Within this food service segment, restaurants, defined in this study as commercial food establishments providing food and beverage services to customers including dine-in, fast food, takeaway, and café-style services, but excluding institutional catering such as schools and hospitals, represent a high-impact but under-researched component.

Restaurants account for a substantial urban carbon footprint, especially during energy-intensive processes, including transportation, storage, heating, cooling, lighting, preparation, and emissions from waste management (Özgen et al. 2021). Primary GHGs associated with restaurant operations and their respective global warming potentials (GWPs) are depicted in Fig. 1. CO_2 from combustion processes, such as cooking and transportation, has a GWP of 1 and is the baseline for comparison. Methane (CH_4), with a GWP of 28, arises from the decomposition of organic waste. Nitrous oxide (N_2O), with a much higher GWP of 265, is linked to fertilizer application, while Hydrofluorocarbons (HFCs), used in refrigeration, exhibit the highest GWP of 1300, highlighting the critical need for alternative refrigerants and efficient cooling technologies. This information demonstrates the importance of targeting restaurants as essential nodes global carbon reduction strategies.

Restaurant supplies, including food and beverages, constitute about 30% of total tourism expenses (Hu et al. 2013), highlighting the sector's economic and environmental significance. Historically, the food industry's strategies to reduce carbon footprints (CF) prioritized large-scale production and supply chain optimization. However, restaurants have

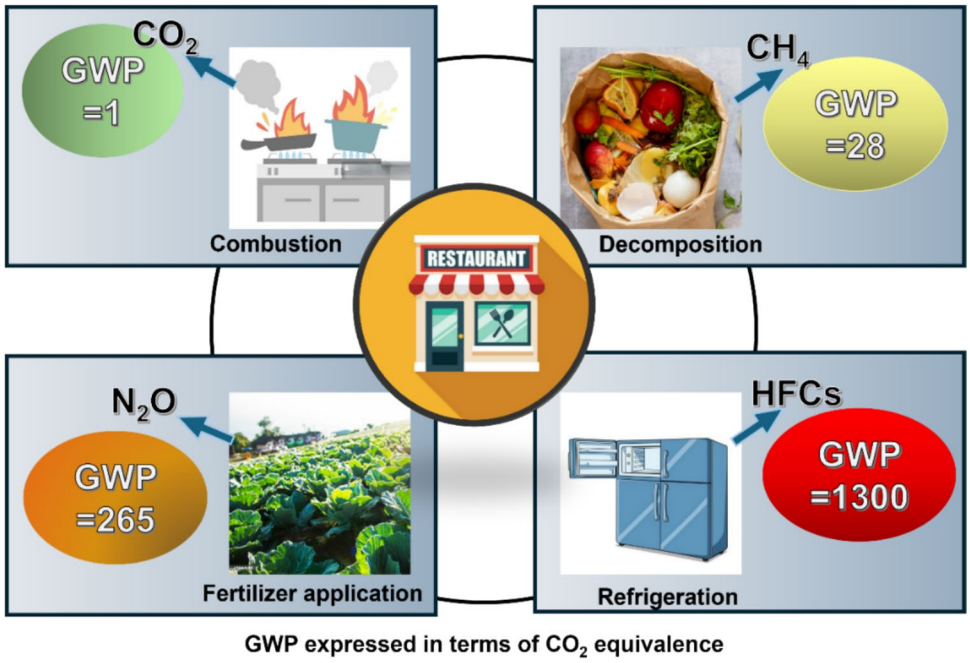

Fig. 1 Common gases emitted in restaurant inventories and their CO_2 equivalence for global warming potentials

a unique potential to influence supply chain practices and consumer behavior. With rising awareness of environmental issues, restaurants are increasingly adopting sustainable practices, such as sourcing local, organic ingredients, minimizing food waste, and integrating energy-efficient technologies (Orynycz et al. 2020; Sha'ari et al. 2023). Structural shiftss, including the transition to renewable energy sources, adopting efficient appliances, and implementing zero-waste food management, offer significant potential for reduction in emissions. For instance, composting and waste-to-energy solutions can mitigate methane emissions from decomposing landfill-bound food waste, as depicted in Fig. 1 (Majumdar et al. 2013).

Advances in sustainable business models, such as carbon offsetting and digital technologies, further support these efforts (Torres et al. 2022). Carbon offsetting allows restaurants to invest in projects like reforestation or renewable energy development to balance their emissions. Simultaneously, digital tools powered by IoT and AI enhance energy tracking, waste management, and inventory optimization (Pagaldiviti and Roy 2023). Despite these developments, challenges persist, including high costs, limited access to advanced technologies, insufficient policy incentives, and a lack of standardized frameworks for achieving carbon neutrality.

The potential for sustainable menus featuring locally sourced and plant-based ingredients has disproportionately lower impacts than animal-based options. However, recent evidence suggests that sustainable meals may also lead to greater food waste due to reduced consumer satisfaction, revealing a potential tension between low-carbon diets and waste reduction goals (Bux et al. 2025). Effective waste management could potentially solve high due to substantial food waste (Sha'ari et al. 2023). Transitioning to renewable energy, such as solar or wind power, along with adopting technologies like solar cookers and alternative refrigerants, can further reduce emissions from energy-intensive operations (Savaşan 2022).

Previous reviews on CF in restaurants have examined the sustainability indicators based on water, energy, gas supply, and waste reduction (Maynard et al. 2021). Elsewhere, sources of environmental unsustainability and green practices adopted to attain zero-net carbon emissions have been explored (Madanaguli et al. 2022). Despite the steady increase in research on restaurant carbon emissions in recent years, there is still a knowledge gap in synthesizing existing literature on strategies, challenges, and innovations for reducing carbon emissions from restaurants. This review addresses the gap by exploring policy frameworks, technological solutions, and case studies of successful low-carbon-emission initiatives within the industry. By consolidating current research and introducing a framework for practical implementation, this study provides actionable insights for stakeholders, policymakers, and restaurateurs. Insights gained herein offer guidance to the restaurant sector toward sustainable practices and reduced environmental impact.

2 Methodology

This study is a systematic literature review to synthesize current knowledge on carbon emission reduction strategies, innovations, and implementation challenges in the restaurant sector. Herein, empirical and theoretical studies relevant to restaurant sustainability practices are evaluated and consolidated. By applying a structured selection and synthesis process, the study offers a comprehensive overview of current approaches and research gaps in the field.

2.1 Literature search and selection criteria

A comprehensive literature search was conducted across Web of Science (WoS) and Google Scholar to identify high-quality, peer-reviewed studies addressing sustainable restaurant practices,

energy efficiency, and carbon footprint mitigation. The search covered literature published between January 2015 and December 2024, using Boolean operators and targeted keyword combinations including: "carbon footprint" AND restaurant, "carbon emissions" AND restaurant, and "low-carbon strategies" AND restaurant. The inclusion criteria required that the peer-reviewed studies focus specifically on GHG emissions, energy use, waste management, or carbon mitigation strategies in restaurant contexts, and present empirical findings, case analyses, or policy frameworks. Exclusion criteria eliminated studies that addressed only agricultural or domestic cooking practices, lacked relevance to environmental sustainability, were non-peer-reviewed, or lacked substantive methodological or outcome-based data. The initial search yielded 92 records, which were reduced to 89 unique entries after duplicate removal. Titles and abstracts were screened for relevance, resulting in 51 full-text articles assessed for eligibility. An initial screening was then conducted based on the content of the abstracts. After applying the predefined inclusion and exclusion criteria, 33 studies were retained for final synthesis. The selection and screening process is visually summarized in the PRISMA flow diagram Fig. 2.

Data extraction was conducted with a focus on thematic relevance, targeting key dimensions within each selected study. Specifically, extracted data included the type of intervention including, food waste reduction, energy efficiency, and sustainable sourcing, the geographical context, the method of greenhouse gas (GHG) estimation or measurement, and any reported outcomes or implementation challenges. A thematic synthesis was subsequently performed

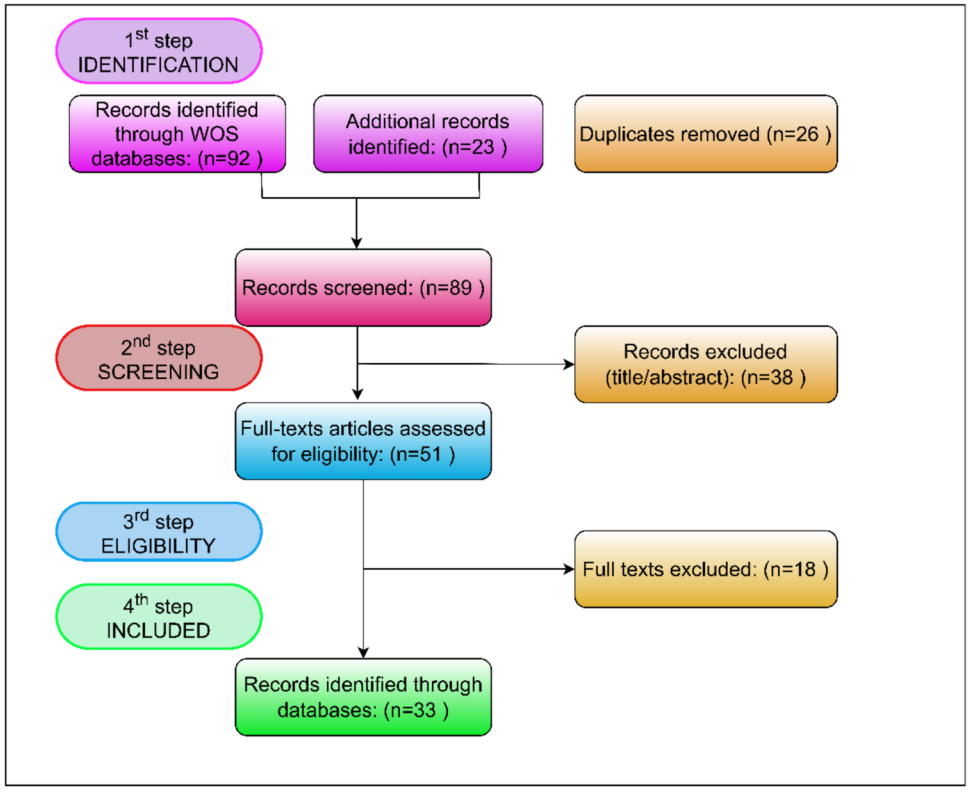

Fig. 2 PRISMA flow diagram for the systematic review illustrating the literature multi-step filtering process, including identification, screening, eligibility assessment, and inclusion

to identify prevailing research directions, expose critical knowledge gaps, and derive actionable insights to inform low-carbon transitions in restaurant operations.

2.2 Thematic Dimensions and Analytical Frameworks for Carbon Footprint Assessment

To support the objectives of this review, data extraction and analysis were guided by key thematic dimensions identified within the selected studies. These included the type of sustainability intervention implemented, such as food waste reduction, energy efficiency technologies, sustainable sourcing, the geographic context of the study, the GHG estimation method or life cycle assessment (LCA) framework employed, and the reported outcomes and implementation challenges. Both quantitative methods, such as emission inventories, LCA, and utility data analysis, and qualitative approaches, including case studies, material flow analysis, and interviews, were used to ensure comprehensive coverage of the literature. This review adopts a holistic framework for analyzing restaurant carbon emissions, structured around the internationally recognized classification of Scope 1, Scope 2, and Scope 3 emissions (Messier 2016; Mungkung et al. 2024). Scope 1 includes direct emissions from on-site fuel combustion used in cooking and space heating, refrigerant leakage, and restaurant-owned delivery vehicles. Scope 2 emissions encompass indirect emissions from purchased electricity for lighting, HVAC systems, refrigeration, and other appliances. Scope 3 emissions, which typically constitute the largest share of a restaurant's total carbon footprint, cover upstream and downstream emissions across the value chain. These include food production, packaging, third-party logistics, customer transportation, packaging disposal, and post-consumer product use.

Several studies reviewed employed established analytical tools, including the IPCC Tier 1 methodology, ISO 14044-compliant LCA, and emissions databases such as EcoInvent. While these methods provide robust insights into direct and indirect emissions, many studies also emphasized critical limitations, particularly in accurately quantifying Scope 3 emissions. The complexity of global and regional food supply chains, reliance on multiple intermediaries, and a lack of transparent reporting hinder consistent emissions tracking in this category. Furthermore, regional disparities in policy frameworks, economic conditions, and data availability contribute to challenges in standardizing carbon footprint assessments across diverse restaurant settings. The limited adoption of internationally recognized reporting standards, such as ISO 14064 or PAS 2050, further constrains comparability and reliability in carbon reporting. These findings highlight the urgent need for harmonized methodologies, improved supply chain transparency, and enhanced stakeholder collaboration to support credible and actionable carbon footprint assessments within the restaurant sector.

3 Carbon Footprint in Restaurants: Integrating Products, food processing Dynamics, and Meal Assessments

Understanding the research landscape surrounding carbon footprint (CF) reduction in the restaurant sector is essential for identifying progress, gaps, and emerging trends in sustainability science. As the global foodservice industry faces mounting pressure to reduce its environmental impact, academic interest in low-carbon strategies for restaurants has grown considerably.

Bibliometric analysis offers a systematic means of quantifying this scholarly engagement, revealing how research outputs are distributed across time, geography, and thematic focus. This section presents an overview of recent literature trends, highlighting publication volume, leading contributing countries, and areas of concentrated academic activity.

There has been a consistent upward trend in scholarly publications addressing carbon footprint (CF) emissions from the restaurant sector since 2017, culminating in a peak of 27 publications in 2024 Fig. 3a, reflecting increased academic engagement with this area of research. As shown in Fig. 3b, China and the United States are the most prolific contributors to this field. Within Asia, Taiwan ranks third after China and India, followed closely by Japan. Most publications originate from a limited number of high-output countries, whereas several others demonstrate relatively modest research activity, as the color-coded classification indicates. To ensure topical relevance and analytical consistency, the bibliometric analysis focused exclusively on studies published within the last decade, encompassing both quantitative and qualitative investigations related to GHG emissions, energy consumption, and waste management practices in the restaurant industry.

Restaurants are integral to tourism, where food and beverage services account for approximately 30% of expenses and significantly influence supply chain logistics, energy systems, and consumer behavior (Vuksanović et al. 2024). The role of the restaurant and tourism sectors makes them a critical sector for reducing global carbon footprint and advancing environmental sustainability (Han 2021). Key Performance Indicators (KPIs), with standardized metrics to evaluate energy consumption, waste generation, and supply chain emissions, are essential for CF evaluations (Özgen et al. 2021). A comparison of the carbon footprints of restaurants studied in different regions is presented in Table 1. Technological innovations, including digital monitoring systems and Internet of Things (IoT) technologies, are critical for precise energy tracking, inventory optimization, and waste reduction. Artificial intelligence (AI) is helpful for demand forecasting and automated energy management to improve operational efficiency and minimize waste (Torres et al. 2022).

The compilation of empirical studies across various geographical contexts in Table 1 reveals a rich yet fragmented landscape of carbon footprint (CF) quantification and mitigation in the restaurant sector.

Despite differing methodologies, data sources, and scopes of analysis, a set of converging themes and strategic insights emerges. A central observation is the critical role of operational context, including fuel types, ingredient sourcing, and food waste practices, in shaping restaurant CF profiles. For instance, studies from Turkey (Özgen et al. 2021) and India (Majumdar et al. 2013) demonstrate how differences in energy sources fir instance, LPG vs. biomass fuels, substantially alter emission outputs, underscoring the need for region-specific decarbonization pathways. This aligns with findings from Toronto (Wren 2004), where restaurants contribute significantly to urban GHG emissions due to fossil fuel dependence, suggesting the urban restaurant sector as an under-addressed hotspot for city-level climate planning.

Equally important is the dominance of product raw materials, especially meat, as emission drivers, as highlighted in studies from Finland (Byggmästar 2019) and Helsinki (Pulkkinen et al. 2016). Beef alone accounted for over half of total emissions in one case, reinforcing the substantial mitigation potential of menu redesign toward plant-based or lower-emission protein sources. These findings complement emerging concepts such as Climate Choice meals, which use meal-level LCA to guide dietary shifts. Another insight lies in the integration of food waste analytics with CF estimation, particularly in Malaysia Sha'ari et al. 2023) and Indonesia (Fernandez et al. 2024). These studies provide rare quantification of emissions by waste type, preparation, serving, and plate loss, illustrating how targeted interventions for instance, efficient kitchen appliances, food redistribution, or composting) Simultaneously address resource inefficiencies and reduce Scope 3 emissions.

The review also exposes a notable disparity between assessment sophistication and practical application. While advanced tools like SimaPro (Baldwin et al. 2011) and ISO 14044-based LCAs have demonstrated impact in institutional settings such as through GS-46 certification, such frameworks remain largely absent in small to mid-sized establishments. This highlights a critical implementation gap, suggesting the need for simplified, scalable tools tailored for broader sector adoption. Carbon offsetting remains a nascent yet underleveraged strategy, with only a few studies reporting verifiable interventions, such as afforestation (Özgen et al. 2021). The integration of offsetting mechanisms into carbon accounting systems, such as through blockchain traceability or voluntary reporting protocols, could strengthen the credibility and attractiveness of low-carbon transitions in the foodservice industry.

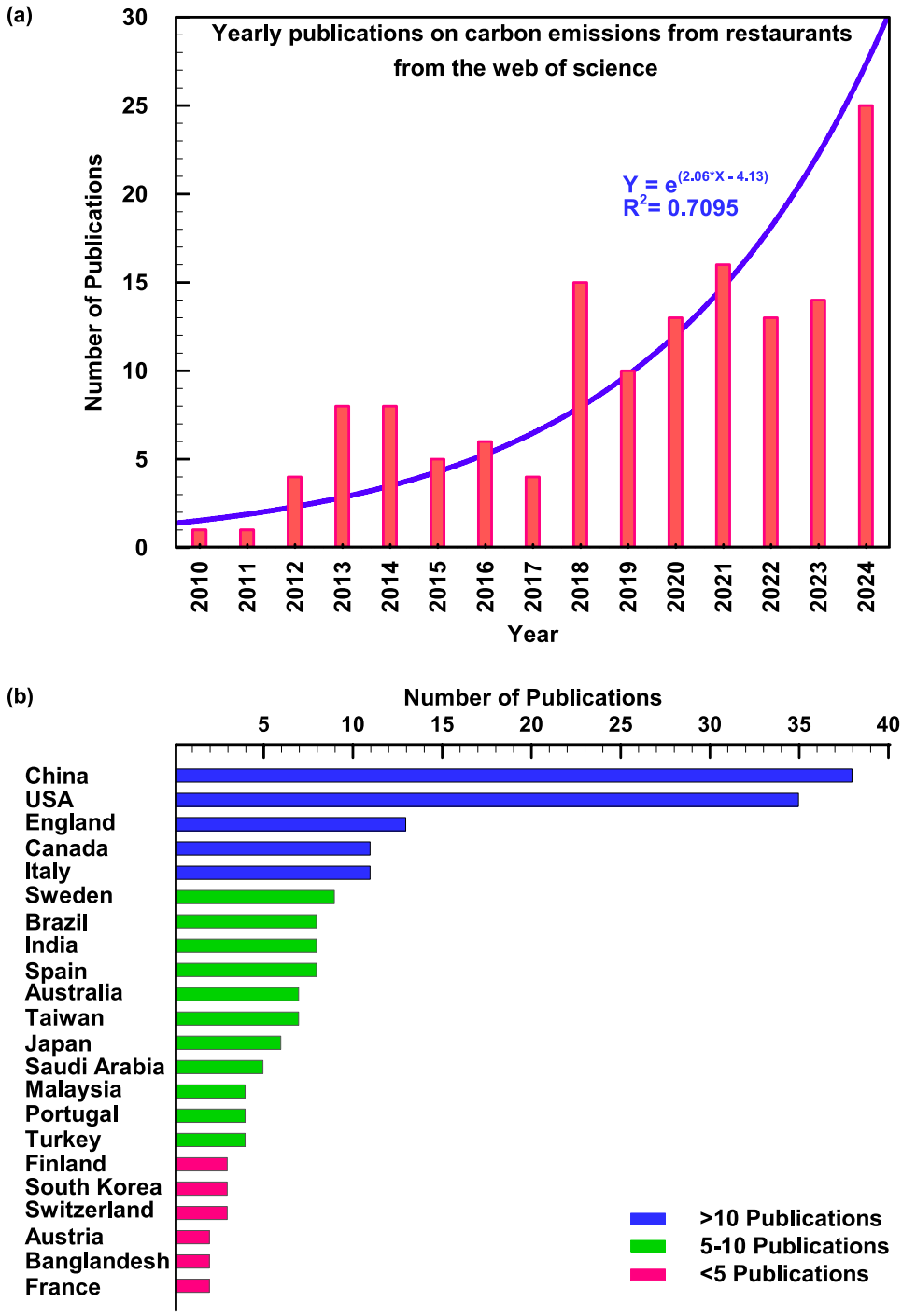

Fig. 3 **a** The evolution of annual scientific publications on carbon emissions from restaurants from 2010 to 2024 and **b** the horizontal bar chart representing the publications on carbon footprints from restaurants by countries. From the Web of Science

Taken together, these studies affirm the multi-source nature of emissions in restaurant operations, while pointing to actionable levers: fuel switching, ingredient substitution, waste management, and operational efficiency. Importantly, they suggest that the most effective decarbonization strategies are not isolated, but rather interlinked across supply chains, consumer behavior, and technological systems. Unlocking these synergies requires a shift from fragmented carbon assessments toward holistic, systems-based frameworks that integrate real-time monitoring, standard metrics, and cross-sector collaboration.

Notable results include substantial GHG reductions through Green Seal Standards in the USA, high emissions linked to specific practices such as, wood use in Indian cities, and the impact of meal composition and food waste on carbon footprints globally.

Table 1 Summary of recent investigations on carbon footprint in restaurants

Location of study	Method	Key Variables	Carbon Footprint	Carbon offsetting Techniques	Reference
Local Pizza restaurant in Izmir, Turkey	Tier 1 methodology (IPCC)	Utility bills and purchasing records (LPG and electricity)	1,919.8 tCO2e for 5 years	Afforestation -3062 seedlings	(Özgen et al. 2021)
Urban and sub-urban areas of Malaysia	Material flow analysis	Food waste: Preparation loss, serving loss, and customer's plate loss	Preparation loss (51.4%), serving loss (30.95%), and customer's plate loss (17.8%) Total average CF: Electricity (13.9–29.5 $KgCO_2e$), water consumption 5.9–22.9 $KgCO_2e$	Application of waste as livestock feed and adopting kitchen equipment with higher energy efficiency	(Sha'ari et al. 2023)
USA	Development of Green Seal Standard (GS-46) LCA using SimaPro and ISO 14044 guidelines	food procurement, storage, preparation, service/support, plastic disposal	GHG reduction of 80.9 t CO_2e annually for restaurants that implemented GS-46 standard	Responsible food purchases,	(Baldwin et al. 2011)
Finnish restaurant	Product raw materials	Product raw materials Operational emissions	1611 tCO_2e, product raw materials (76.2%), beef (55%)	Meal adjustments	(Byggmästar 2019)
Indian cities of Nagpur and Raipur	Per capita emissions		Adjusting menu order to prioritize low-carbon options can significantly reduce the average meal carbon footprint	Raipur's CF high (overreliance on wood and candy coal) Nagpur's CF is low ()	(Majumdar et al. 2013)
Toronto	Proportion of city's total GHGs	Fuels	2.4% of the city's total GHGs	–	(Wren 2004)
Helsinki, Finland	LCA for 105 common meals	carbon footprints per meal	1.2 $kgCO_2e$	–	(Pulkkinen et al. 2016)
Indonesia, Makassar, Ternate and Bogor cities	Material flow analysis	Food waste	46.69–62.28 kgCO2eq/plate/year		(Fernandez et al. 2024)

CO_2e stands for carbon dioxide equivalent

3.1 Carbon Footprint in Product Raw Materials and Operational Emissions

The literature reveals a strong reliance on both quantitative and qualitative approaches to assess carbon footprint (CF) in the restaurant sector. Quantitative methods, such as emission inventories, life cycle assessment (LCA), and utility-based audits, are widely used to capture emissions across Scopes 1, 2, and 3. These allow for the detailed quantification of emissions from direct sources such as fuel combustion, refrigerant leakage. and indirect sources such as purchased electricity and supply chain activities. Complementing these, qualitative analyses, including case studies and thematic reviews, help contextualize emission drivers within behavioral, managerial, and policy frameworks.

Mixed-methods studies provide a more integrated perspective. For example, Özgen et al. (2021) applied the IPCC Tier 1 approach to assess CF in a Turkish pizza chain, reporting that LPG alone contributed 916.47 tCO_2e out of a total 1,919.82 tCO_2e over five years. Similarly, Sha'ari et al. (2023) used a four-phase methodology to evaluate food waste and CF in Malaysian casual dining restaurants, noting regional differences in energy and water usage that significantly influenced overall emissions.

In addition to operational emissions, raw material sourcing emerged as a dominant emission hotspot. A Finnish restaurant chain attributed 76.2% of its 2018 emissions (1,611 tCO_2e) to food ingredients, particularly beef, which accounted for 55% of the total (Byggmästar 2019). Consistent with this, Pulkkinen et al. (2016) demonstrated that plant-based dishes such as lentils, tofu, beans generated substantially lower CFs than those containing meat or energy-intensive dairy products such as cheese.

Fuel choice also significantly affects operational emissions. In Indian cities, restaurants using biochar, wood, and coal emitted more CH_4 and N_2O than those using LPG (Majumdar et al. 2013). Regional differences were evident: Raipur's heavy reliance on solid fuels led to higher per capita emissions compared to Nagpur. In Toronto, restaurants accounted for approximately 2.4% of citywide GHG emissions, with natural gas being the primary source (Wren 2004). These findings underscore the influence of local energy sources, fuel types, and policy contexts on CF outcomes.

Beyond carbon emissions, the reviewed studies also link restaurant operations to environmental and public health impacts**.** Food procurement alone contributed over 50% of climate change impacts and 97% of land use pressure in U.S. foodservice operations (Baldwin et al. 2011).. Cooking processes also emit harmful volatile organic compounds (VOCs); in Hong Kong, commercial kitchens were found to release substantial quantities of formaldehyde and long-chain carbonyls, contributing to indoor air pollution and necessitating energy-intensive ventilation (Ho et al. 2006; Chen et al. 2021).

These findings indicate the need for multi-level, region-specific strategies to reduce emissions. Effective interventions include transitioning to cleaner fuels, enhancing energy efficiency, improving ventilation infrastructure, and adopting low-carbon menus. Additionally, policy instruments, such as sustainability certifications, subsidies, and carbon reporting mandates, could incentivize broader adoption of climate-conscious practices across the restaurant industry.

3.2 Life Cycle Assessments and CF Analysis of Menus and Operations

Across the reviewed literature, Life Cycle Assessment (LCA) consistently emerged as the most widely employed analytical framework for evaluating the environmental impacts of restaurant operations. Rooted in ISO 14044 standards, LCA facilitates a systematized evaluation of emissions across multiple stages of the restaurant value chain, from food procurement and storage to preparation, service, and waste disposal (Baldwin et al. 2011). Software tools such as SimaPro are commonly used to implement this approach, while complementary methodologies like Material Flow Analysis (MFA) have proven valuable for tracking food waste at critical stages, including preparation, serving, and post-consumer disposal, key contributors to Scope 3 emissions.

Despite its methodological robustness, the broader adoption of international standards such as ISO 14064 (for GHG quantification and reporting) and PAS 2050 (for product-level carbon footprint assessment) remains limited within the restaurant sector. This is particularly evident among small and medium-sized enterprises (SMEs), which often lack the technical capacity and resources to integrate these frameworks into routine operations.

Recent studies have expanded the application of LCA to assess not only operations but also the carbon intensity of specific menu items and food sourcing strategies. In a case study conducted at a university dining service in Harrisonburg, Virginia, (Striebig et al. 2019) applied LCA to evaluate the secondary emissions associated with sourcing local versus non-local ingredients. The results indicated that items such as lettuce and strawberries transported from California had up to four times the emissions of locally sourced alternatives. A modest 10% increase in local sourcing yielded a reduction of over 4000 kg CO_2-equivalent annually. Similarly, in Finland, local ingredient sourcing was found to reduce transportation-related emissions to just 0.3% of total CF (Byggmästar 2019). emphasizing the environmental advantages of regional supply chains.

The concept of Climate Choice meals, as proposed by (Pulkkinen et al. 2016), offers an additional lens for integrating LCA findings into consumer-facing strategies. Using simplified LCA for 105 common meals, the study found that carbon footprints ranged from 0.6 to 2.8 kg CO_2e per meal, with an average of 1.2 kg CO_2e. Notably, beef-based items such as the Classic Burger (2.34 kg CO_2e) were significantly more carbon-intensive than chicken (0.96 kg CO_2e) or plant-based options like the Vegan Burger (0.50 kg CO_2e). Based on this, the Climate Choice framework recommended emissions thresholds for menu items, defined as 25% below the average meal CF, to qualify as low-carbon choices. Vegetarian and non-endangered fish-based meals often met these criteria, reinforcing the mitigation potential of menu-level interventions.

Collectively, these findings underscore the utility of LCA and related environmental assessment tools in identifying high-impact ingredients, supply chain inefficiencies, and operational hotspots. Moreover, the integration of carbon footprint information into menu design and sourcing decisions, supported by standard-compliant assessment protocols, can play a pivotal role in reducing emissions while enhancing transparency and consumer engagement in sustainable dining practices.

3.3 Carbon Emissions From Restaurants' Food and Non-Food Wastes

Restaurants encounter significant challenges associated with managing food and non-food waste, both of which contribute to environmental degradation and carbon emissions. Food waste arises from poor forecasting, inefficient cooking practices, and consumer behavior, with preparation losses accounting for the highest proportion at 51.37%, followed by serving loss (30.95%) and plate waste (17.68%) in Malaysian restaurants (Sha'ari et al. 2023). Non-food waste includes plastics, inefficient energy use, and excessive water consumption, further exacerbating sustainability concerns. Waste management accounts for 0.8% of total emissions, underscoring the importance of targeted interventions to minimize waste (Byggmästar 2019).

Waste cooking oil presents an opportunity for sustainable biodiesel production, offering dual benefits of reducing environmental contamination and providing an alternative to fossil fuels. Uncollected waste oil contributes significantly to soil and water pollution; however, its conversion to biodiesel improves fuel properties, engine performance, and emission profiles (Onuh et al. 2021). In New Zealand's hospitality sector, 24,375 tonnes of food waste are generated annually, with 61% deemed avoidable CF (Mazzetto et al. 2023). Notably, food waste contributes less than 1% to the overall carbon footprint, yet preparation waste (60%) and plate waste (33%) remain primary contributors, driven by inaccurate demand forecasting, oversized portions, and inadequate staff training (Dhir et al. 2020). These findings emphasize the need for robust inventory management, portion control, and waste reduction strategies to address overproduction and minimize the environmental impact of waste in the restaurant sector.

Recently, digital technologies and artificial intelligence (AI) have been highlighted as effective tools for CF redustion in restaurants by enabling precise measurement and monitoring of food waste. AI-driven inventory management, predictive demand modeling, and real-time monitoring systems are implemented to minimize overproduction and spoilage (Bux 2024). Specifically, AI-integrated 3D scanners and digital scales help classify food categories, quantify waste, and evaluate associated environmental impacts. These tools support data-driven decision-making, contributing to more sustainable food service practices and reduced emissions.

3.4 Key Contributions and Research Directions for Carbon Neutrality in Restaurants

Restaurants contribute to environmental degradation through food and non-food waste, and high energy and water consumption. Addressing inefficiencies in forecasting, cooking, and packaging, alongside adopting green practices like portion control, reusable materials, and local sourcing, can significantly reduce their carbon footprint. As seen in Fig. 4, previous research highlights key areas of research impact and high activity, reflecting fields with significant scientific, societal, or technological importance. Figure 4a illustrates meso-topics frequently discussed in publications related to carbon footprints in restaurants, categorized based on broader research focus themes. The most significant areas, such as Environmental Sciences (41) and Sustainability Science (25), emphasize the dominant attention on environmental impacts and strategies for achieving sustainability in the restaurant industry. Other significant mesotopics include Agricultural Policy (18), highlighting the role of farming practices and food sourcing, and Combustion (14), reflecting emissions from cooking processes. Topics like Social Psychology (14) and Bioengineering (8) demonstrate the field's interdisciplinary nature, addressing behavioral influences and technological innovations for mitigation. Smaller but noteworthy areas, such as Hospitality & Tourism (4), Supply Chain (4), and Toxic Agrochemicals (3), highlight specialized research on sector-specific emissions and impacts. Together, these meso-topics provide a comprehensive overview of the multifaceted approaches and research directions aimed at reducing the carbon footprint of restaurants.

Figure 4b provides a detailed visualization of micro-topics frequently explored in publications related to carbon footprints in the restaurant industry. Each micro-topic represents a specific area of focus, such as energy consumption, waste

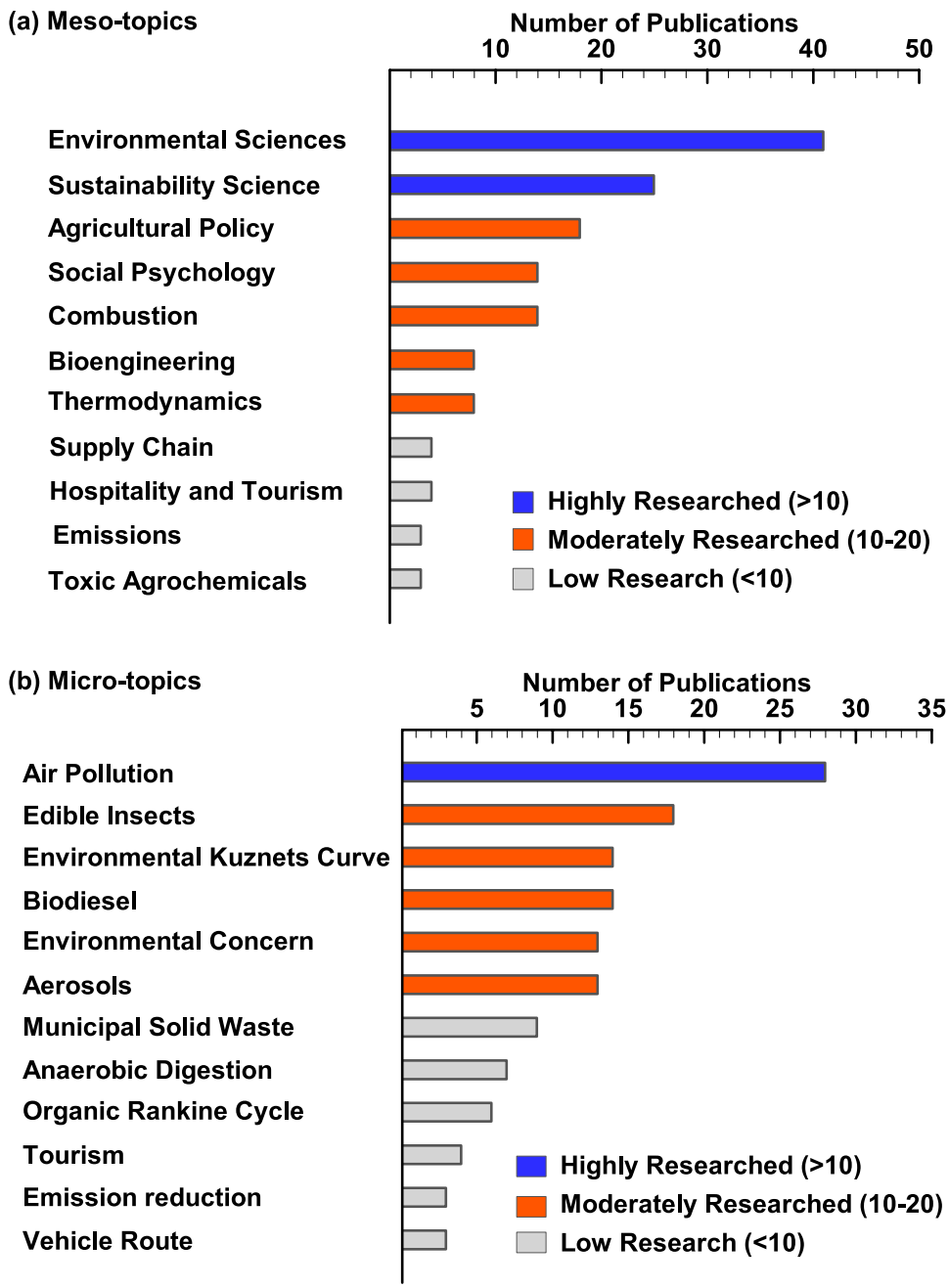

Fig. 4 Citation based classification of published literature on CF in restaurants **a** micro-topics and **b** meso-topics

management, supply chain emissions, and the environmental impact of cooking practices. The prominence of each topic reflects its relative importance or the extent of its exploration in existing literature. For instance, topics like food procurement, which significantly contribute to land use and climate change impacts, are central to understanding the sector's overall carbon footprint. Similarly, emissions from cooking processes, such as the release of carbonyl compounds, highlight the role of operational practices in air pollution. The multidimensional nature of restaurant-related emissions and the need for targeted strategies addressing these diverse but interconnected aspects to achieve decarbonization goals are emphasized.

Standardized KPIs for energy use, food waste, and emissions are crucial for tracking progress, while waste management, water conservation, and supply chain strategies offer practical solutions to minimize resource use and emissions. For instance, the Green Restaurants ASSessment (GRASS) tool is a validated instrument for evaluating restaurant sustainability (Maynard et al. 2021). Compared to 97 restaurants in Brazil, all CF elements linked to restaurants are classified into three sections: water, energy, and gas supply; menu and food waste; waste reduction; construction

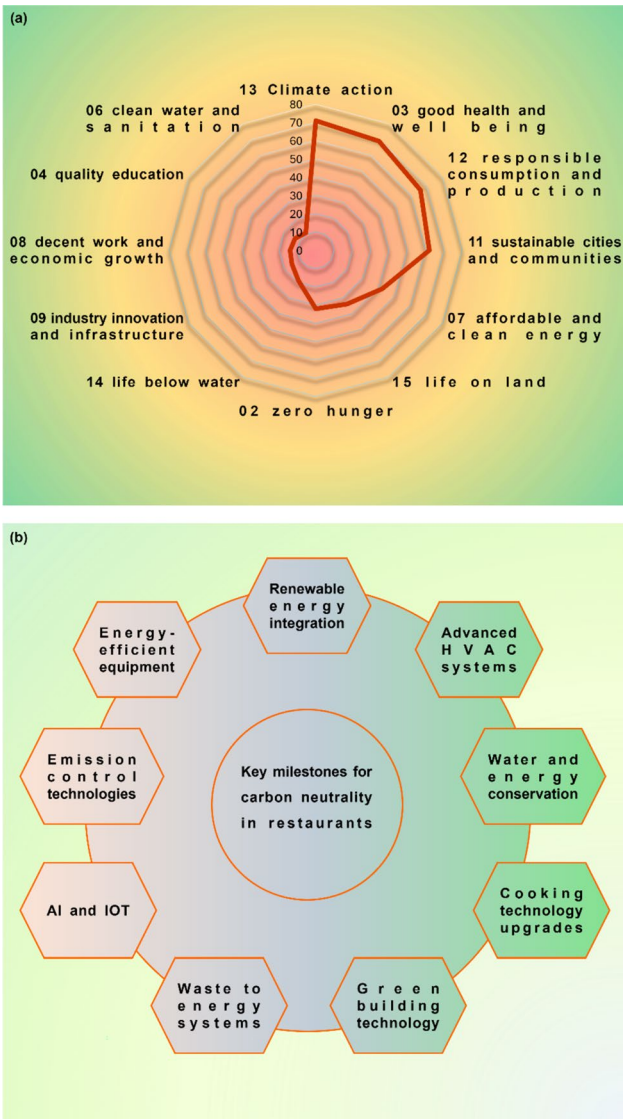

Fig. 5 **a** A radar plot representing the contributions of low-carbon technologies in restaurants to attaining sustainable development goals, and **b** Key milestones for achieving carbon neutrality in restaurants, highlighting nine key innovative technological and operational milestones across energy, waste, and resource management

materials; and social sustainability. Scores are categorized as green (≥ 75%), yellow (> 40%– < 75%), and red (≤ 40%). None of the restaurants achieved green classification, while 47.4% received a yellow rating, primarily due to gaps in greenhouse gas reduction and locally sourced food practices. Previously, quantitative analysis revealed significant disparities in sustainability efforts as summarized in other previous investigations (Madanaguli et al. 2022).

Recent publications have highlighted key milestones in green restaurant ecosystem frameworks integrating stakeholders, sustainability processes, and outcomes (Özgen et al. 2021). Energy-efficient equipment and advanced HVAC systems optimize energy consumption, while renewable energy integration and water conservation practices significantly lower greenhouse gas emissions from utilities. Cooking technology upgrades enhance efficiency and reduce emissions during food preparation, while waste-to-energy systems repurpose organic waste, minimizing methane emissions from decomposition. Green building technologies promote sustainable infrastructure, and emission control technologies tackle pollutants from cooking and energy use. Incorporating AI and IoT facilitates real-time monitoring and optimization of energy and waste management. Collectively, these measures target the primary sources of carbon emissions in restaurants, aligning operations with sustainability goals and advancing carbon neutrality.

The contributions of low-carbon restaurant technologies to the United Nations Sustainable Development Goals (SDGs) are presented in Fig. 5a. The radar chart highlights the extent of impact across various goals, with the strongest contributions observed in 13: Climate Action (Torres et al. 2022), reflecting efforts to mitigate greenhouse gas emissions and 03: Good Health and Well-Being, indicating improved air quality and reduced health risks from pollution (Mutuku et al. 2020). Significant contributions are also noted for 12: Responsible Consumption and Production, driven by efficient resource use and waste reduction, and 11: Sustainable Cities and Communities, showcasing advancements in urban sustainability through clean energy and efficient infrastructure.

Moderate impacts are visible in goals such as 07: Affordable and Clean Energy, emphasizing the role of renewable energy technologies, and 06: Clean Water and Sanitation, reflecting the role of sustainable practices in water conservation. However, the chart indicates minimal contributions to goals like 02: Zero Hunger and 14: Life Below Water, where the influence of low-carbon technologies is less direct. The concentric gradient from green to red in the background reinforces the relative intensity of contributions, with green representing a more substantial alignment with specific SDGs. Overall, low-carbon technologies address diverse sustainability aspects, Fig. 5b. The areas with limited contributions to carbon neutrality, such as clean water and sanitation, quality education, decent work and economic growth, industry innovation and infrastructure, life below water, and zero hunger, are prominently highlighted in Fig. 5a.

4 Advancing Carbon Neutrality Framework in Restaurants: Challenges and Strategic Solutions

While the number of studies on carbon footprint (CF) in restaurants has grown, significant methodological and operational challenges persist, particularly in the accurate quantification of Scope 3 emissions. The inherent complexity of restaurant supply chains, involving multiple suppliers, intermediaries, and processing steps, complicates the tracking of upstream emissions related to food production, packaging, and distribution. Furthermore, disparities in regional infrastructure, policies, and economic conditions create inconsistencies that hinder the standardization of assessment methodologies across geographies.

A major limitation remains the lack of integration of formal environmental management systems within restaurant operations. Most establishments do not align with global standards such as ISO 14064 or PAS 2050, resulting in fragmented reporting practices and poor data comparability. Data collection frequently relies on self-reported inputs such as energy, water, and material use gathered through questionnaires, site visits, or interviews, with supporting background data sourced from databases like EcoInvent. However, these sources vary in scope, depth, and reliability. Studies employing tools such as Eco-indicator 99 reveal a broad spectrum of environmental impacts, including land use, fossil fuel consumption, and respiratory inorganics, but underscore the need for improved consistency and harmonization across datasets and reporting protocols.

In response to these limitations, there is a growing recognition of the need for standardized key performance indicators (KPIs), digital tools, and sector-specific accounting frameworks to support robust CF evaluation. The absence of such structures continues to impede industry-wide benchmarking and policy alignment. Moreover, many sustainability efforts remain superficial or fragmented, hindered by limited consumer awareness, logistical constraints in waste handling, and variability in restaurant size and service models (Pulkkinen et al. 2016; Özgen et al. 2021).

To advance carbon neutrality, several technological and operational interventions have been proposed. Transitioning to low-carbon fuels, such as biogas, biodiesel, ethanol, and hydrogen, has demonstrated substantial mitigation potential. For instance, biogas derived from organic waste can simultaneously address emissions and food waste, while biodiesel and bioethanol offer reductions of up to 86% compared to petroleum-based fuels (Foteinis et al. 2020; Amalia et al. 2021). Waste-to-energy systems, including the conversion of used cooking oil and paper waste into renewable energy, have been shown to produce between 13.9 and 17.2 million gigajoules annually in the UK's quick-service restaurant sector, highlighting the scalability of such solutions.

Operational strategies further include menu redesign to favor low-emission ingredients such as, poultry, legumes, and plant-based proteins, adoption of energy-efficient appliances, low-flow water systems, and heat recovery technologies. Scenario modeling suggests that comprehensive adoption of these interventions could reduce operational emissions by up to 64%. In parallel, certifications such as the Green Seal GS-46 standard and LEED frameworks promote best practices in energy management, procurement, and waste reduction (Baldwin et al. 2011). Nature-based offsets, such as planting Pinus brutia seedlings, can also complement technology-driven strategies, particularly in regions with afforestation potential (Özgen et al. 2021).

The integration of circular economy principles is increasingly regarded as essential. Practices such as redistributing surplus food, batch preparation, and inventory optimization help reduce food waste and resource inefficiencies (Sha'ari et al. 2023). At a systems level, the adoption of standardized carbon tracking tools, transparency mechanisms, and cross-sector collaboration is critical to enabling consistency and scalability in carbon mitigation efforts.

Finally, future progress will depend heavily on digital innovations. Technologies such as the Internet of Things (IoT) and artificial intelligence (AI) have already shown promise in optimizing energy consumption, predicting waste patterns, and enhancing real-time monitoring. When combined with regulatory support and industry-wide cooperation, such tools can drive the restaurant sector toward a more decarbonized and resilient future.

In alignment with these strategies, Fig. 6 provides a detailed analysis of greenhouse gas (GHG) emissions associated with food inputs, meal preparation, and dietary patterns across diverse regions. Figure 6a examines the GHG emissions per kilogram of raw food materials used in restaurants. It highlights the disproportionately high emissions of animal-based foods such as beef, lamb, and lobster, with beef alone contributing over 25 kg CO_2e per kilogram (Hu et al. 2013; Özgen et al. 2021). By contrast, plant-based foods like vegetables, fruits, and grains generate significantly lower emissions, often below 2 kg CO_2e per kilogram (Baldwin et al. 2011). This contrast emphasizes the need to shift toward plant-based alternatives to mitigate the environmental impact of food systems. Building on this, Fig. 6b depicts an evaluation of GHG emissions per serving for various meal options, distinguishing between emissions from material inputs and food processing. High-emission meals, such as the double bacon burger and classic burger, derive most of their emissions from beef patties, showcasing the environmental cost of beef-heavy menus (Goldstein

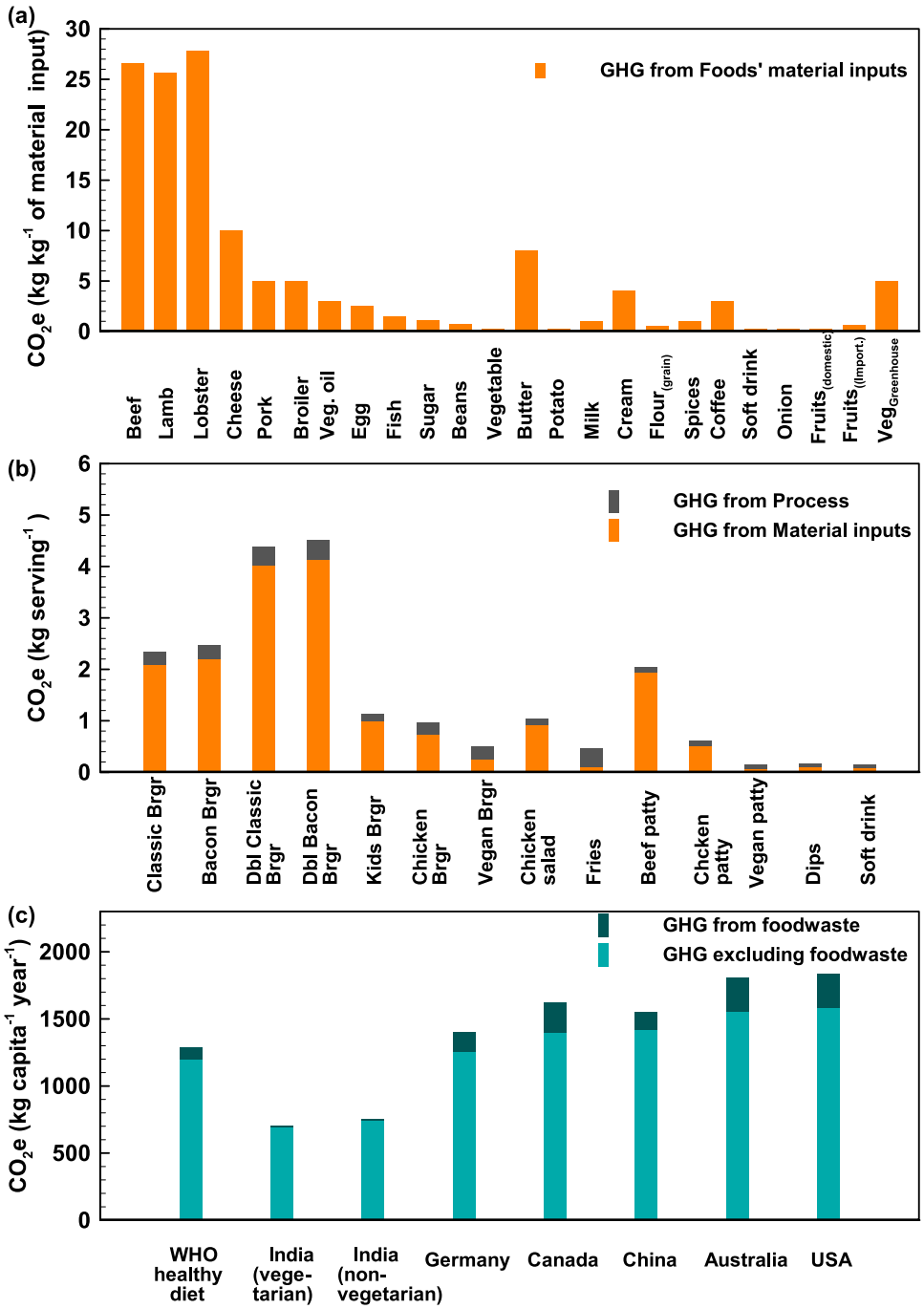

Fig. 6 GHG emissions across the food lifecycle: **a** GHG from material inputs for various restaurant foods (Clune et al. 2017), **b** GHG emissions from ingredients and processing of different restaurant menu items (Byggmästar 2019), and **c** Annual per capita GHG contributions from diets, highlighting the role of food wastes in different countries (Ritchie et al. 2018)

et al. 2017; Babakhani et al. 2020). Conversely, plant-based options like the vegan burger and chicken-based meals, such as chicken salad, demonstrate significantly lower emissions. These findings reinforce the potential of incorporating low-impact ingredients to reduce emissions in the restaurant industry.

Further analysis depicted in Fig. 6c compares per capita GHG emissions from diets across countries, factoring in the role of food waste (Striebig et al. 2019). The United States, Australia, and China lead in per capita emissions, particularly when food waste is included. In contrast, diets with lower meat consumption, such as India's vegetarian diet and the WHO-recommended healthy diet, exhibit significantly lower per capita emissions. These results underscore the critical role of dietary choices and effective food waste management in reducing carbon footprints globally. Overall, the results illustrate the interconnected factors driving GHG emissions in the restaurant and food service industry. The sector can play a pivotal role in advancing sustainability and achieving significant reductions in

greenhouse gas emissions by addressing the environmental impact of food inputs, optimizing processing methods, and promoting dietary shifts alongside waste management strategies (Fig. 6).

5 Conclusions and Outlook

This review highlights the significant contribution of the restaurant industry to carbon emissions, primarily through fuel combustion, energy use, and supply chain activities, with Scope 3 emissions dominating its carbon footprint. Sustainable practices, including transitioning to green fuels like biogas, ethanol, and hydrogen, adopting second-generation biodiesel, and integrating circular economy principles, significantly reduce greenhouse gas emissions. Operational improvements, including energy-efficient appliances, heat recovery systems, and low-carbon menu redesigns, could potentially cut emissions by up to 64%. Standardized Key Performance Indicators for energy, waste, and emissions are essential for tracking progress and harmonizing carbon footprint assessments across diverse restaurant types. Waste-to-energy systems and composting offer practical solutions to minimize food and non-food waste, while nature-based interventions, such as planting Pinus Brutia seedlings in Turkey, demonstrate scalable mitigation strategies. Frameworks like the GS-46 standard emphasize waste audits, energy management, and responsible procurement to foster sustainability. This review underscores the integrated collaborative policies promoting renewable energy and harmonized methodologies for addressing data gaps and supporting the sector's transition to carbon neutrality.

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Declarations

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